Roll No. | | | | | | | | | | | | Total No. of Pages :07

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B.Pharmacy (Sem.-1)

HUMAN ANATOMY AND PHYSIOLOGY-I

Subject Code : BP/101T

M.Code : 74644

Date of Examination : 12-06-2024

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TWENTY questions carrying ONE mark each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Answer briefly :

   a. The extracellular fluid compartment consists of which of the following?

      i) Vascular and transcellular

      ii) Interstitial, vascular and connective tissue fluid

      iii) Intra-cellular and transcellular

      iv) Trans-cellular, intra-cellular and connective tissue fluid.

   b. What is the movement of water molecules across a plasma membrane from the side where the solution concentration is more dilute to the side where the solution is more concentrated called?

      i) Osmosis

      ii) reverse osmosis

      iii) diffusion

      iv) hydration

c. Diffusion is the term given to the process where :

i) molecules move along their concentration gradient from high concentration to low concentration.

ii) water moves along its concentration gradient from low concentration to high concentration.

iii) ATP is used to move ions along their concentration gradient.

iv) A membrane protein, by changing shape after binding to a molecule, moves the molecule across the plasma membrane.

d. A hypertonic solution is one which :

i) has an osmotic pressure that is different to that inside red blood cells.

ii) has an osmolarity less than that of red blood cells.

iii) causes no net movement of water through the membrane of red blood cells.

iv) has an osmolarity greater than that of red blood cells.

e. If a patient was suffering from "acidosis", what would this mean?

i) Blood pH is not sufficiently alkaline.

ii) Blood pH is acidic.

iii) There is too little hydronium ion in the plasma.

iv) Blood pH is too acidic.

f. To which of the following class of biological compounds do all enzymes belong?

i) Hormones

ii) Proteins

iii) Carbohydrates

iv) Lipids.

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g. Which glands secrete "oil" into a hair follicle?

i) Apocrine

ii) Eccrine

iii) ceruminous

iv) sebaceous.

h. The human body's ability to maintain a relatively constant internal temperature is an example of what?

i) Respiratory heat loss

ii) Homeostasis

iii) Vasodilation and evaporative heat loss

iv) Positive feedback.

i. Which bone of the head has a synovial joint?

i) The sphenoid

ii) The maxilla

iii) The mandible

iv) The hyoid.

j. Which of the following is the smallest structure within a muscle fibre?

i) Myosin

ii) Myofilament

iii) Myofibril

iv Sarcomere.

k. What are red blood cells also known as?

i) Erythrocytes

ii) Thrombocytes

iii) Monocytes

iv) Eosinophils.

l. What name is used for a nerve cell?

i) Neuron

ii) Neuroglia

iii) Ganglion

iv) Astrocyte

m. What part of the nervous system prepares the body for action during extreme situations?

i) The limbic system

ii) The sympathetic division

iii) The efferent system

iv) The parasympathetic division

n. What is the range of frequencies that the human ear is most sensitive to?

i) 50Hz to 500Hz

ii) 12,000 Hz to 20,000 JHz

iii) 500Hz to 6000Hz

iv) 20 Hz to 20,000 Hz.

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o. What is the study of how body parts function called?

i) Histology

ii) Physiology

iii) Homeostasis

iv) Metabolism.

p. In which part of a cell does the process of making ATP from oxygen and glucose take place?

i) Lysosomes

ii) Ribosomes

iii) Mitochondria

iv) Golgi apparatus.

q. Which of the following is the smallest living structural unit of the body?

i) Atom

ii) Molecule

iii) Organelle

iv) Cell.

r. Which layer of the skin is the most superficial?

i) Epidermis

ii) Dermis

iii) Papillary dermal layer

iv) Stratum germinativum.

s. Number of facial bones :

i) 9

ii) 5

iii) 12

iv) 14.

t. Normal blood pressure of human body

i) 70/130 mmHg

ii) 80/120 mmHg

iii) 90/110 mmHg

iv) 60/150 mmHg.

SECTION-B

2. Draw a well labelled diagram of eye and describe its functions.

3. Define cell and tissues. Classify different types of tissues. Write structure and functions of epithelial tissues.

4. What is blood? Write different types of blood cells. Explain any two blood disorder.

SECTION-C

5. Write a short note on cellular signalling.

6. Explain in detail nerve action potential.

7. Define coagulation. Explain detailed coagulation mechanism.

8. Write the significance of Reticulo endothelial system.

9. Explain functions of the lymphatic system.

10. Write the differences between sympathetic and parasympathetic nervous system.

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